HW4

25. $\int x^3 e^{x^2}\,dx$

Try:

① $\begin{cases} u = x^3 \\ dv = e^{x^2}\,dx \end{cases}$ $\begin{cases} du = 3x^2 \\ v = \int e^{x^2}\,dx \end{cases}$

$= ?$

NOT obvious how to move forward

② $\begin{cases} u = e^{x^2} \\ dv = x^3\,dx \end{cases} \Rightarrow \begin{cases} du = e^{x^2} \cdot 2x\,dx \\ v = \frac{x^4}{4} \end{cases}$

$\downarrow$

$\int v\,du = \int e^{x^2} \cdot 2x \cdot \frac{x^4}{4}\,dx$

$= \int e^{x^2} \cdot \frac{x^5}{2}\,dx$

$\rightarrow$ not obvious

Let's try Substitution.

25. $\int x^3 e^{x^2}\,dx$

$$I = \int x \cdot x^2 \cdot e^{x^2}\,dx$$

$$u = x^2 \Rightarrow du = 2x\,dx$$

$$\Rightarrow \frac{du}{2} = x\,dx$$

$$I = \int x^2 e^{x^2} \cdot x\,dx = \int u e^u \cdot \frac{du}{2}$$

$$\Rightarrow I = \frac{1}{2}\int u e^u\,du$$

then use integration by parts for this

Similarly:

$$\int x^7 \cdot e^{x^2}\,dx$$

$$\int x^6 e^{x^2}\, x\,dx$$

$$u = x^2 \rightarrow \frac{1}{2}\int u^3 \cdot e^u\,du$$

tabular method

LIATE : Set u be one of the below function

L : log : Example $\ln x$; $\ln(x+1)$; $\log(x)$ ....

I : Inverse Trig : Example : $\sin^{-1}(x)$, $\tan^{-1}(3x+1)$ ....

A : Algebraic : Ex : x^2 , x, $(x^3 + x)$ ....

T : Trig Ex : $\sin x$, $\cos x$ .....

E : Exponential Ex : e^x, e^{2x+1} , 2^x ....

Example :

$$\int x \cdot \sin x \, dx$$

A (Algebraic) T (Trig)

$\begin{cases} u = x \\ dv = \sin x \, dx \end{cases}$ b/c A comes before T in LIATE

$$\int x \sin^{-1} x \, dx$$

Algebraic Inverse Trig

$$\begin{cases} u = \sin^{-1} x \\ dv = x\,dx \end{cases}$$

b/c I comes before A in LIATE

⊛ Integration by parts of Definite integral.

$$\int_a^b u \cdot dv = u \cdot v \Big]_a^b - \int_a^b v \cdot du$$

Example:

$$I = \int_{-1}^{1} \ln(x+2)\,dx$$

$$\begin{cases} u = \ln(x+2) \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \frac{1}{x+2}\,dx \\ v = x \end{cases}$$

$$\Rightarrow I = u \cdot v \Big|_{-1}^{1} - \int_{-1}^{1} v \cdot du$$

$$\Rightarrow I = x \cdot \ln(x+2) \Big|_{-1}^{1} - \int_{-1}^{1} \frac{x}{x+2} \cdot dx$$

$$I = \ln 3 - \underbrace{\int_{-1}^{1} \frac{x}{x+2}\, dx}_{J}$$

$$J = \int_{-1}^{1} \frac{x}{x+2} \cdot dx$$

$$= \int_{-1}^{1} \frac{(x+2)-2}{x+2}\, dx$$

$$= \int_{-1}^{1} \left(\frac{x+2}{x+2} - \frac{2}{x+2} \right) dx$$

$$= \int_{-1}^{1} \left(1 - \frac{2}{x+2} \right) dx = \int_{-1}^{1} dx - \int_{-1}^{1} \frac{2}{x+2}\, dx$$

$$= x\Big|_{-1}^{1} - 2 \cdot \ln(x+2)\Big|_{-1}^{1}$$

$$= 2 - 2(\ln 3)$$

$$I = \ln 3 - (2 - 2\ln 3)$$

$$I = 3\ln 3 - 2$$

29. $\int_1^e x^2 \ln x \, dx$